WATERLOO ROCKETRY

Next-Gen Ground Support Software System (Orion)

Preliminary Design Review

Qian Qian
Software Lead

Shrey Shingala
Software Lead

October 29, 2026

Agenda

01 Overview, Motivation & Feasibility

02 Requirements

03 Explanation of Key Technological Choices

04 Data Modeling

05 Deployment Architecture

06 Testing

07 Project Management

08 Action Items

SECTION 01

Overview, Motivation & Feasibility

01

WATERLOO ROCKETRY

Overview

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.

Motivation & Feasibility

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.

SECTION 02

Requirements

02

WATERLOO ROCKETRY

Requirements Overview

The GitHub issues `2026-2027-software-issues#51` contains and is the root parent of issues containing:

- general technical directions
- design (tasks and decisions)
  - specification
  - API Protocol and Data Model
- implementation (requirements and tasks)

Please review its content.

General Requirements

Requirement	Motivation
Omnibus MUST remain production-ready until Orion is fully production-ready and is fully replacing Omnibus.	Allow smooth transition and ensure fallback option exists while issues are ironed out.
Orion SHOULD provide one authoritative API server for telemetry, shared state and command routing.	Provides a stable boundary between sources and consumers, and avoid system-wide breaks when a source or consumer is updated.
Public interfaces SHOULD be typed, versioned, documented and machine-checked.	Standardized interfaces to ensure safe interaction and validation of data.
Complete and well organized documentation MUST be provided for all public interfaces, including API, data model, and configuration.	Ensure that the system is usable and mainytainable by future developers and users.
The deployment of the system MUST be reproducible.	Ensure that the system can be deployed in a consistent manner.
The system MUST be easy to configure and the configuration methods MUST be documented.	Ensure that the system can be configured by all users. (Abandoning the current Python-script-based configuration system.)

SECTION 03

Explanation of Key Technological Choices

03

WATERLOO ROCKETRY

SECTION 04

Data Modeling

04

WATERLOO ROCKETRY

SECTION 05

Deployment Architecture

05

WATERLOO ROCKETRY

SECTION 06

Testing

06

WATERLOO ROCKETRY

SECTION 07

Project Management

07

WATERLOO ROCKETRY

Timelines, and Reviews

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do.

SECTION 08

Action Items

08

WATERLOO ROCKETRY